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An investigation of PPP-B2b coverage and its performance in ZTD estimation and positioning in different regions

Xiaoming Wang^{id}*, Kai Zhou^a, Jinglei Zhang^a, Haobo Li^c, Hong Liang^d, Manhong Tu^d, Yufei Chen^{a,b}, Cong Qiu^{a,b} and Li Li^e

This study comprehensively evaluates the Beidou navigation satellite system's PPP-B2b service for real-time positioning and zenith total delay (ZTD) estimation. Despite its operational status, PPP-b2b's effective coverage is underexplored. We define this coverage by analysing satellite visibility and geometry and evaluating ZTD and positioning accuracy at 48 stations. Results show PPP-B2b achieves ZTD Errors of 10–20 mm in China and 15–30 mm in Europe, with kinematic experiments in China and Japan showing consistent accuracy within 20 mm. This study highlights PPP-b2b's potential for enhancing atmospheric monitoring and precise positioning in its effective regions.

Keywords: BDS-3, PPP-B2b service, GNSS, PDOP, Zenith total delay, Positioning, PPP

Introduction

Global navigation satellite systems (GNSS), which are designed mainly for positioning, navigation, and timing services (Leick et al., 2015), have been used to remotely sense the Earth's atmosphere since the 1990s. The precise point positioning (PPP) technique is widely used in estimating the zenith total delay (ZTD), which can be converted to precipitable water vapor (PWV) using surface temperature and pressure measurements at the GNSS site. However, the accuracy of ZTD derived using the PPP technique is highly dependent on the quality of the satellite orbit and clock data, as these errors are introduced into the PPP processing and are neither eliminated nor accounted for (Zhou et al., 2020) with the PPP service via the B2b signal (PPP-B2b) of the Beidou Satellite System (BDS) and the high-accuracy service (HAS) of Galileo becoming operational in recent years, high-accuracy orbit and clock corrections are now available through these basic navigation constellations. Before the availability of PPP-B2b service provided by BDS and the HAS from Galileo, we usually needed to use the real-time service (RTS) initiated by the International GNSS Service (IGS) to access GNSS orbit and clock

correction information in real-time for various applications with PPP (Hadas And Bosy, 2015, Kazmierski et al., 2020). Previous studies have shown that the accuracy of real-time tropospheric estimates using the static PPP method with real-time satellite orbit and clock products can reach 1–2 mm for PWV (Li et al., 2015) and approximately 5–18 mm for ZTD (Hadas et al., 2020). In addition to estimating tropospheric information with fixed GNSS stations using the static PPP method, various studies have indicated that tropospheric information can be monitored using GNSS observations on moving platforms such as ships (Wang et al., 2019), Buoys (Chadwell And Bock, 2001), trains (Aichinger-Rosenberger et al., 2021), cars (Gratton et al., 2021), and aeroplanes (Skone et al., 2006), using the kinematic PPP method. However, these studies have highlighted that the major issue for real-time GNSS PWV retrieval on moving platforms is the limitation of Internet communication. Therefore, sensing ZTD/PWV with either static or kinematic PPP using the IGS RTS service is constrained by the need for a stable Internet connection to receive clock and orbit corrections.

Both BDS and Galileo, through their basic navigation constellations, provide accurate satellite orbit and clock information in real-time, enabling high-accuracy positioning and ZTD estimation using the PPP Algorithm without the need for Internet access. The Galileo aims to provide global high-accuracy positioning with errors of less than 20 cm (95%) and 40 m (95%) in the horizontal and vertical domains, respectively, transmitted through an open format in the E6-B signal (Fernandez-Hernandez et al., 2022, Zhou et al., 2024), the HAS was officially declared to have reached initial service status on January 24, 2023, and previous studies have shown that HAS products can achieve decimeter-level positioning accuracy in

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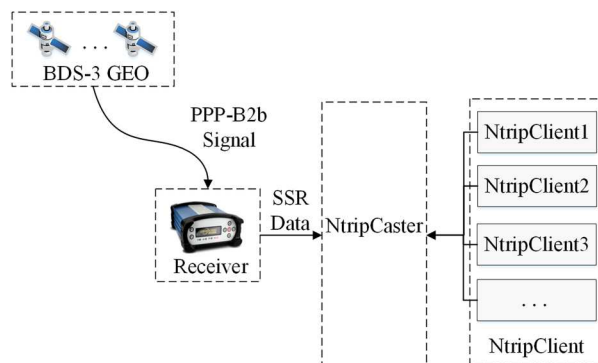
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both static and kinematic modes (Zhou et al., 2024) with the release of the interface control document for PPP service signal PPP-B2b in 2020 (CSNO, 2020), BDS-3 began providing PPP services for China and its surrounding areas via three geostationary Earth orbit (GEO) satellites (Yang et al., 2021). Several recent studies have investigated the performance of PPP-B2b services. Ren et al. (2021) indicated that centimetre-level accuracy can be achieved for both static and kinematic position modes with the PPP-B2b service at the need for Internet access. The high positioning performance of PPP-B2b has been confirmed in several recent studies using GNSS stations located in China (Nie et al., 2021; Tao et al., 2021; Wang et al., 2021; Zhang et al., 2022). Furthermore, applications of PPP-B2b have been extensively studied in various domains, including water vapor monitoring (Yang et al., 2023), time transfer (Ge et al., 2023), and earthquake monitoring (Zang et al., 2024).

However, because PPP-B2b correction information is derived only from the BDS-3 tracking stations deployed in China, its effective service is limited to China and its surrounding areas (Yang et al., 2022). Therefore, the performance of PPP-B2b may vary depending on the number of available satellites and the satellite geometry. Most studies have focused on the performance of PPP-B2b in China, and little attention has been paid to its effective service coverage and performance in different regions, especially outside China to address these shortcomings, we comprehensively investigated the visibility and dilution of precision (DOP) of satellites in terms of PPP-B2b signals in different regions to identify their effective service coverage. The performance of BDS PPP-B2b in positioning and ZTD estimation was evaluated at 48 widely distributed stations using both static and kinematic PPP modes.

Real-time PPP-B2b decoding and transfer via the internet

Because most GNSS receivers cannot be used to decode PPP-B2b signals, we designed a PPP-B2b-Nrtip-Cater system to investigate the performance of real-time positioning and ZTD retrieval in different regions. As illustrated in Figure 1, the PPP-B2b correction information was first received and decoded by a custom-developed receiver located in Beijing, China. These corrections were then sent to our Nrtipcaster Server, which is broadcast using the network transmission of the radio technical



1 Workflow of the PPP-B2b-NRTIP-Cater system.

commission for maritime services under the Internet protocol (Nrtip) and can be accessed via the Internet by any user who would like to investigate the PPP-B2b performance in any region.

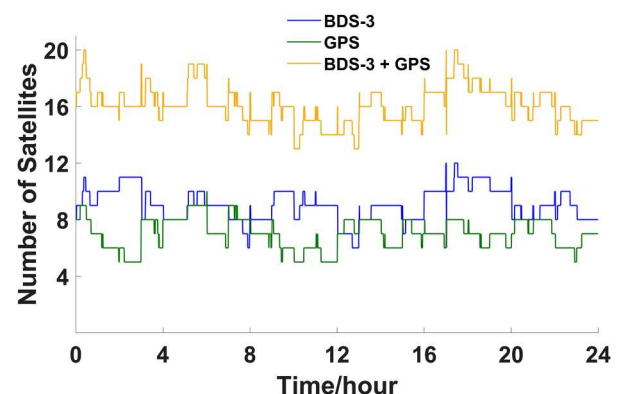
PPP-B2b service availability

Availability is the percentage of time the PPP-B2b service is usable within a specified coverage area. In this study, the availability of the PPP-B2b service is defined as epochs with position DOP (PDOP) ≤ 6 , which is commonly used as a basic service availability threshold with respect to the GPS (Ward, 1997) and BDS-3 performance standards (CSNO, 2018). Therefore, for a region to utilize PPP-B2b service, two criteria must be met: (1) at least one of the three BDS-3 GEO satellites is visible to receive the PPP-B2b correction information; (2) at least four visible GNSS satellites have the correction information from the PPP-B2b signal, and the corresponding PDOP does not exceed six. The availability of four satellites with PDOP values smaller than six is a basic criterion for standard point positioning, which does not ensure that a high-accuracy positioning result can be achieved with PPP. However, we can still provide a preliminary definition of PPP-B2b service coverage using these criteria, and a more comprehensive investigation will be conducted using observations from widely distributed GNSS stations. In this study, the cut-off elevation of the GNSS and BDS-3 GEO satellites was set to 7° .

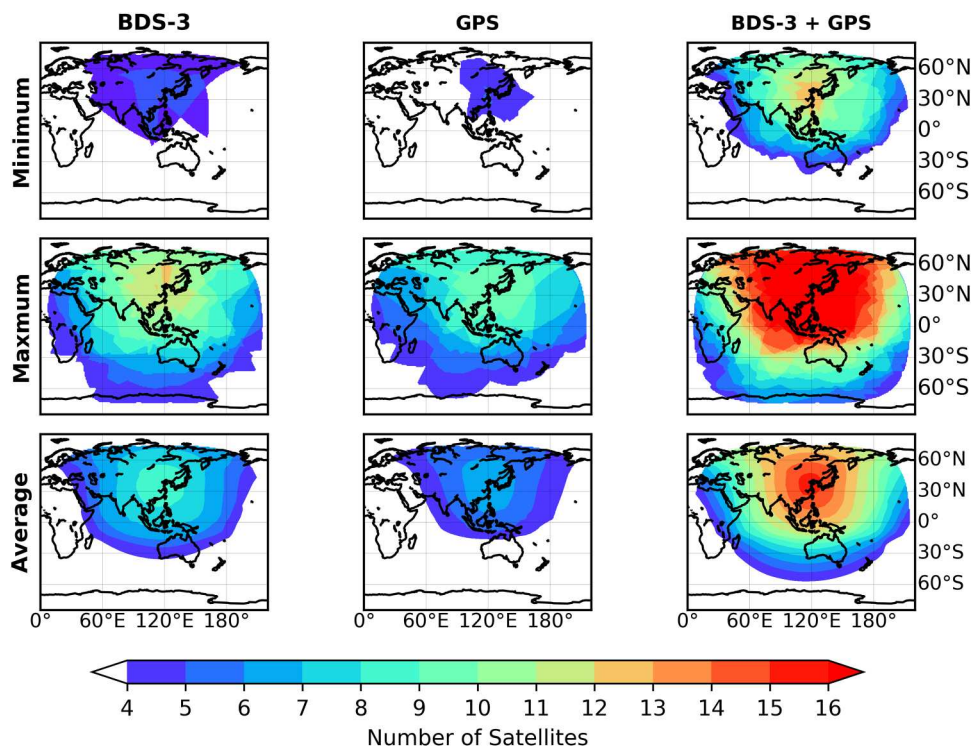
We used the saved PPP-B2b orbit information from July 21, 2022, to investigate the number of visible satellites and corresponding DOP values with a spatial resolution of $1^\circ \times 1^\circ$ and assumed that the altitudes were zero. This allowed us to investigate the minimum, maximum, and average numbers of visible satellites and the DOP values at each grid point, providing us with insights into the availability of PPP-B2b services in different regions.

Number of satellites

Figure 2 shows the number of satellites the PPP-B2b service on July 21, 2022 as observed at the BJ01 site in Beijing, China. On average, nine BDS-3 satellites and seven GPS satellites are available at each epoch of the PPP-B2b service. For most epochs, the number of BDS-3 satellites was



2 Number of satellites for BDS-3 (blue line), GPS (green line), and a combination of BDS-3 and GPS (yellow line) from 00:00 to 24:00 UTC on July 21, 2022, observed at station BJ01 located in Beijing, China.

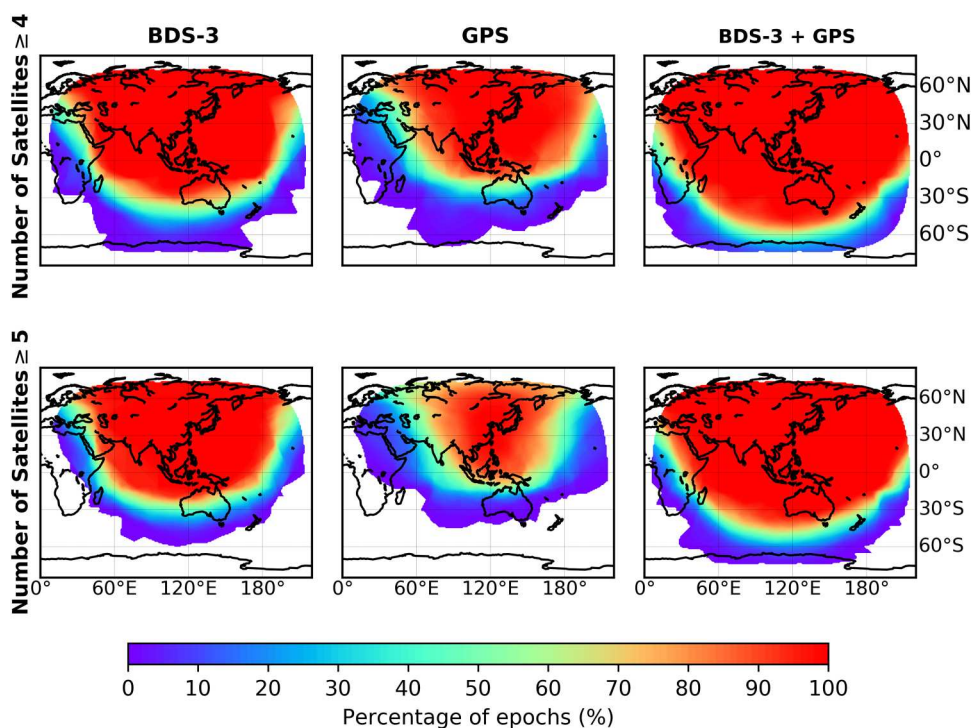


3 Distribution of the minimum (top), maximum (middle), and average (bottom) number of satellites for BDS-3 (left), GPS (middle), and a combination of BDS-3 and GPS (right) in different regions. The number of satellites is represented by different colours, as indicated by the colour bar, with values smaller than 4 marked in white and values larger than 10 marked in red.

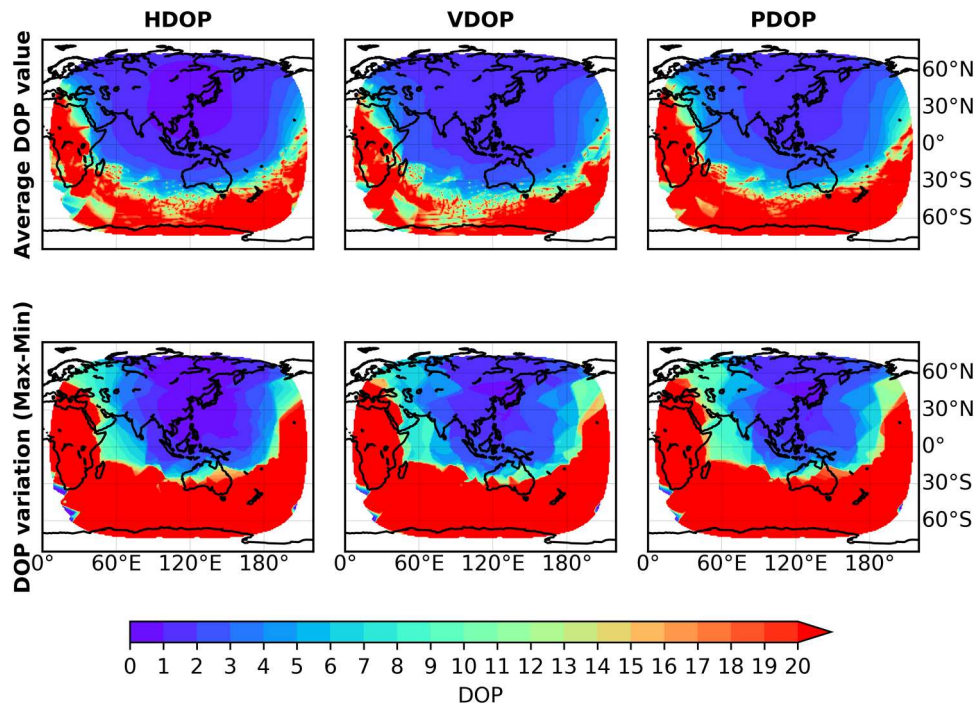
greater than that of GPS satellites. Given the combination of BDS-3 and GPS satellites, the total number of satellites ranged from 13 to 20, with a mean value of 16.

Figure 3 shows the minimum, maximum, and average number of GNSS satellites in different regions. The results

indicate that BDS-3 has slightly better coverage than that of GPS because the number of BDS-3 satellites broadcast in PPP-B2b was greater than that of GPS satellites. The use of a combination of BDS-3 and GPS satellites ensures that at least four satellites are visible in most Asia-Pacific

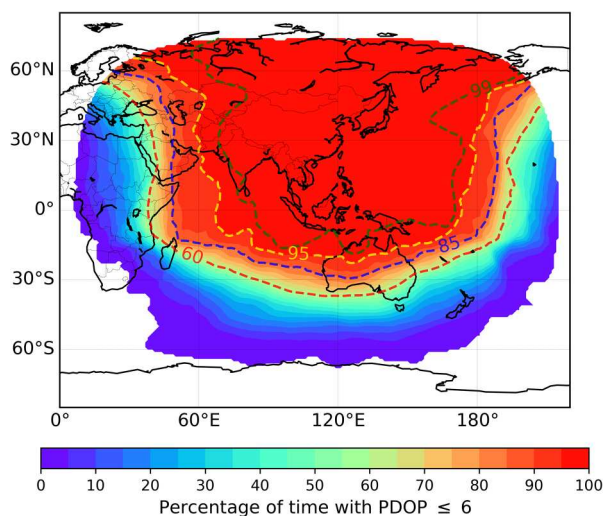


4 Distribution of the percentage of epochs with no fewer than four (top) or five (bottom) visible satellites for BDS-3 (left), GPS (middle), and the combination of BDS-3 and GPS (right). The percentages are represented by different colours, as indicated by the colour bar.



5 Distribution of the average value (top) and variation (bottom) of HDOP (left), VDOP (middle), and PDOP (right). The DOP values are represented by different colours, with values larger than 20 marked in red, as indicated by the colour bar.

countries, Russia, Eastern Europe, and Northeast Africa. The PPP-B2b service provides the most visible satellites for East China, with relatively even coverage across Asia. The average number of satellites (BDS-3 + GPS) steadily decreased from approximately 15 in East China to four in the regions south of Australia and Northeast Africa. Figure 4 shows the percentage of epochs with more than four or five visible GNSS satellites. With a combination of BDS-3 and GPS satellites, at almost every epoch, there are at least four satellites across most Asia-Pacific countries, Eastern Europe, Northeast Africa, the Indian Ocean, and the Pacific Ocean.



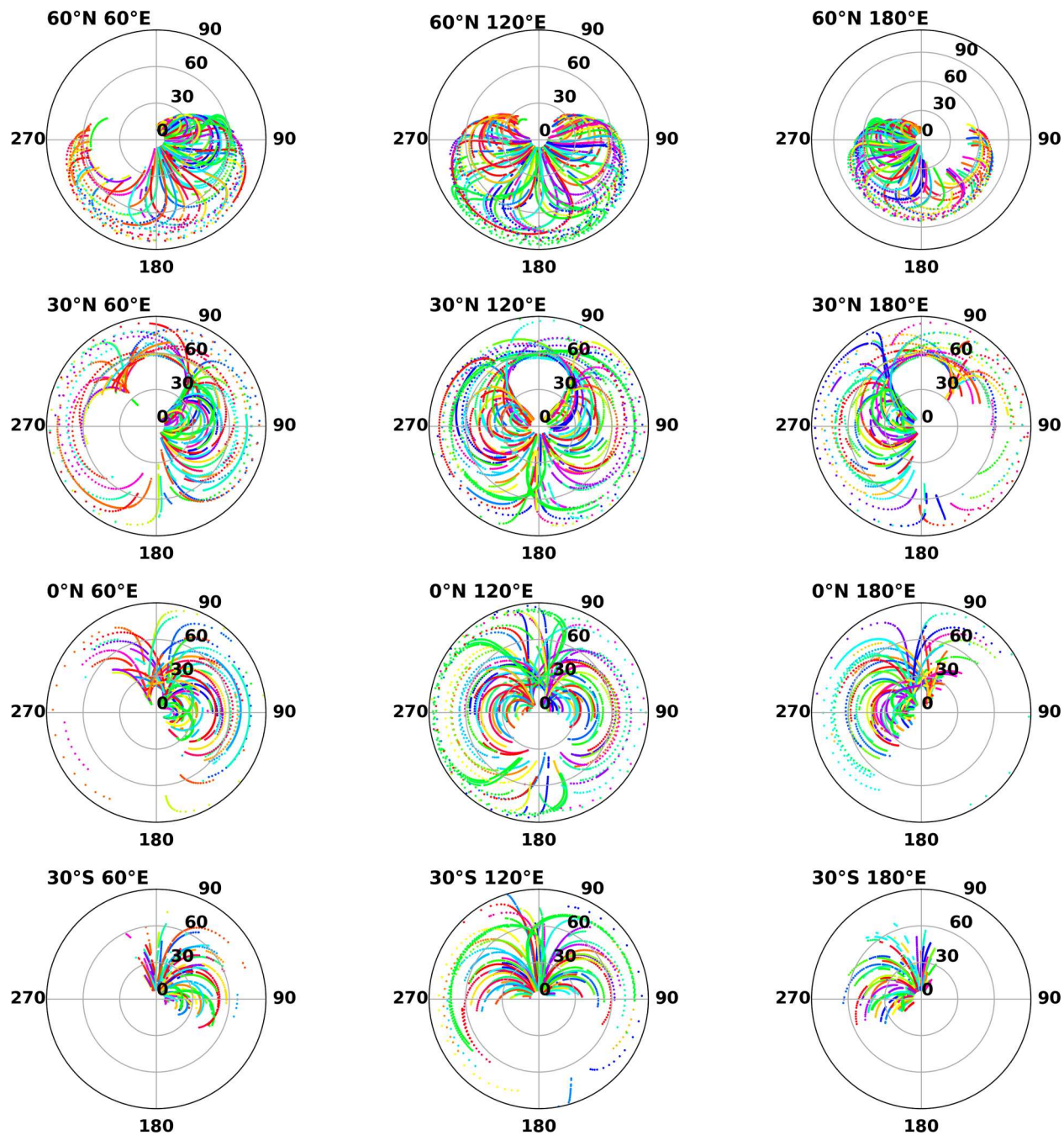
6 Distribution of the percentage of time that $PDOP \leq 6$. The green, yellow, blue, and red dashed lines represent the locations with $PDOP \leq 6$ for 99%, 95%, 85%, and 60% of the time, respectively.

Satellite DOP

In addition to the number of satellites, the DOP, which represents the quality of the geometry of the satellite constellation, is another important indicator of GNSS accuracy. Figure 5 shows the spatial distributions of the average horizontal DOP (HDOP), vertical DOP (VDOP), and PDOP values and their temporal variations (the maximum value minus the minimum value). In line with the spatial distribution of visible GNSS satellites, the average DOP was the smallest (<2) in East China and increased to more than 10 in regions far from China, such as South Australia and Africa. The temporal variation in DOP values in China was less than approximately 5, whereas, for South Australia, the variation was greater than 20.

According to the BDS open service performance standard (CSNO, 2018), the global PDOP availability should meet the requirement of $PDOP \leq 6$ for 95% of the time. Therefore, to identify the PPP-B2b service availability based on this criterion, we calculated the percentage of the time that the $PDOP \leq 6$, and the results are shown in Figure 6. The yellow, blue, and red lines in Figure 6 depict the regions with a $PDOP \leq 6$ for 95%, 85%, and 60% of the time, respectively. Hereafter, we define the effective PPP-B2b service region as the area with a $PDOP \leq 6$, 95% of the time. Figure 6 shows that, compared to the Northern Hemisphere, only a limited region in the Southern Hemisphere was covered by PPP-B2b.

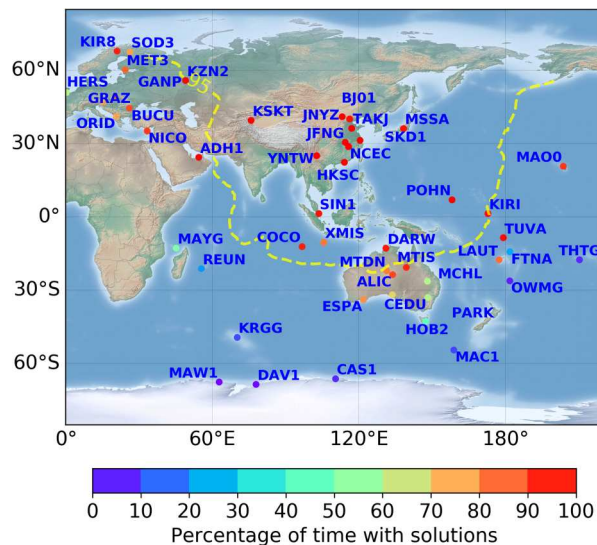
This only provides us with a preliminary definition of PPP-B2b service coverage because its performance in positioning and related applications varies with location due to the differences in the number and geometry of usable satellites. To further illustrate the distribution of usable satellites in the PPP-B2b service for different regions, a sky view plot of all visible satellites on July 21, 2022, from 00 to 24 h UTC at 12 locations is shown in Figure 7. The distribution of usable satellites with



7 Sky view plot of viewable satellites on July 21, 2022, from 00 to 24 h UTC at 12 locations. Different colours represent different GNSS satellites.

Table 1 ZTD estimation strategy with PPP-B2b

Observables	IFLC of pseudorange and carrier phase
Frequencies	GPS L1/L2 and BDS-3 B1I/B3I (B1C/B2a)
Elevation cut-off angle	7°
ZTD estimation	A priori value for the hydrostatic delay, calculated using the Saastamoinen model, and the wet delay was estimated as a random walk process noise of 5 mm/ $\sqrt{h}$
Coordinate	Static mode: coordinates are estimated as constant parameters Kinematic mode: coordinates are estimated as white noise parameters
Mapping function	Global mapping function
Gradient parameters	North and east components of the tropospheric gradient parameters were estimated
Receiver clock	Receiver clock estimated as white noise parameter for GPS and BDS-3 separately
Solution type	Static/kinematic PPP with float ambiguities
Corrections models	Phase wind-up, relativistic delays, the effects of the solid Earth pole tide and the ocean pole tide are modelled according to the IERS Conventions 2010



8 Integrity of the PPP-B2b solution at 48 sites. The yellow line represents the locations with PDOP ≤ 6 for 95% of the time. The PPP-B2b solution integrity is defined as the ratio between the number of PPP-B2b solutions and the number of GBM solutions.

respect to azimuth and elevation varies with location. For example, although the average PDOP values at (60° N, 120° E) and (30° N, 120° E) are both less than two, the satellite tracks can cover almost all azimuths and elevations at (30° N and 120° E), whereas at (60° N, 120° E), the visible satellites are mostly in the southern direction, which may bias the ZTD estimation results if there is a strong atmospheric horizontal gradient.

ZTD retrieval and positioning performance of PPP-B2b

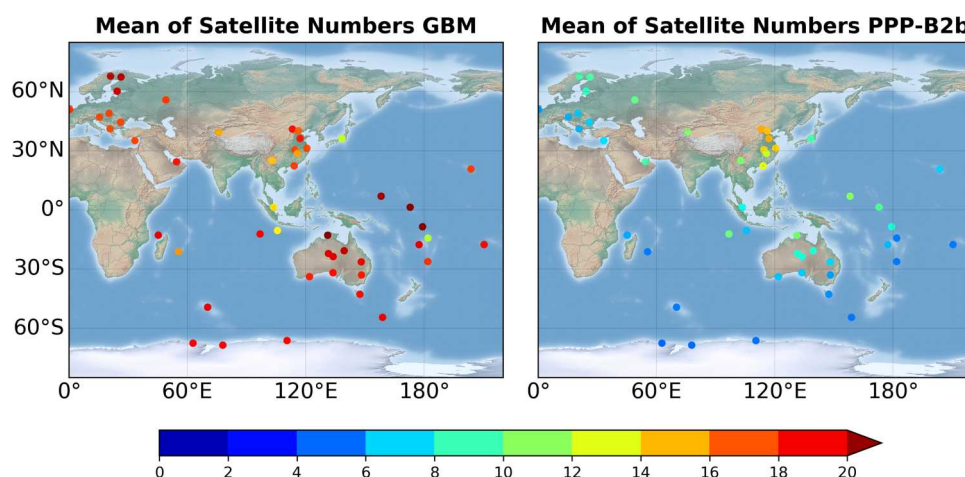
As depicted in the previous section, the DOP, number of satellites, and sky view of the viewable satellites have an obvious dependence on location. Therefore, to investigate how this affects ZTD retrieval and positioning accuracy in different regions, we selected 48 stations for our real-time positioning and ZTD retrieval experiments.

Further details regarding the 48 GNSS stations are provided in the Appendix. With the PPP-B2b information transferred by the system through the Internet, as illustrated in Section 2, and the real-time observation streams at these stations, the ZTD and position were calculated using the PPP technique with 30 s sampling in real-time. The performance of the ZTD and positioning accuracy were then evaluated by comparing them with the post-processed results with Geodetic Benchmark (GBM) final products provided by the Helmholtz Centre Potsdam and German Research Centre for Geosciences. The key processing strategies are listed in Table 1.

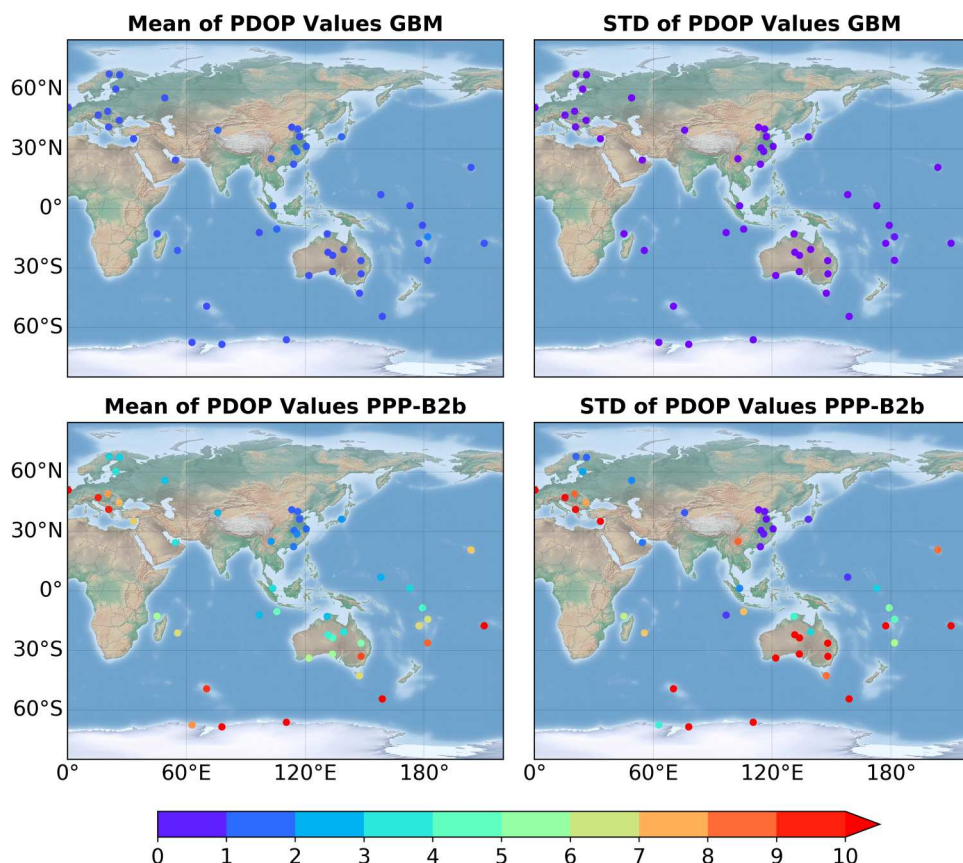
Solution integrity

The accuracy and integrity of the solutions are two important indicators of the PPP-B2b service, which indicate whether this technique can continuously provide users with a reliable positioning service. An insufficient number of visible satellites or poor satellite geometry at certain epochs may lead to a failure in PPP parameter estimation with PPP-B2b corrections, thus affecting the continuity of the PPP-B2b solutions. To investigate the integrity of the PPP-B2b solution, we compared the number of PPP solutions obtained with the PPP-B2b and GBM products. The PPP-B2b solution integrity was defined as the ratio of the number of PPP solutions obtained with the PPP-B2b and GBM products. The integrity of the PPP-B2b solution was defined as the ratio of the number of PPP solutions obtained using the PPP-B2b products to the number obtained using the GBM products. As revealed in Figure 8, for the sites located within the PPP-B2b service region (PDOP ≤ 6 for 95% of the time), the integrity of the solution is close to 100%. For sites located outside the PPP-B2b service region, the solution integrity decreased from above 90% to approximately 75% at GRAZ (Austria) and approximately 50% in South Australia. For sites in the Antarctic, the integrity values were all less than 10%, indicating that PPP-B2b was barely useful in this region.

Solution integrity is mainly related to the number and geometry of usable satellites. Typically, it is not possible to obtain a reliable solution using PPP when the number of satellites is very small or the PDOP value is very large.



9 Mean number of visible satellites (cut-off elevation $\geq 7^\circ$) for the GBM and PPP-B2b products between DOY 202 and 208 at 48 sites. The different colours represent the number of mean satellites as indicated by the colour bar, with values larger than 20 marked in red.



10 Mean (left) and STD value (right) of PDOP for GBM and PPP-B2b products between DOY 202 and 208 at 48 sites. The colour bar shows the mean or STD of PDOP values with corresponding colours, where values larger than 10 are marked in red

As shown in Figure 9, when the GBM products were used, the number of usable satellites was greater than 15 at 43 of the 48 stations. The lowest number of usable satellites was observed at the FTNA and MSSA stations (approximately 12), which could only observe a few of the BDS-3 satellites (PRN number <30) owing to the limitations of their receivers. For the PPP-B2b products, the number of usable satellites was greater than 10 at only 13 stations and showed a strong dependence on location. Several sites with receivers cannot track new BDS-3 satellites (usually PRN > 30), which significantly reduces the number of usable GNSS satellites. The statistical results show that the STDs of the number of usable satellites are less than two across all stations, indicating that the number of usable satellites is stable for both GBM and PPP-B2b.

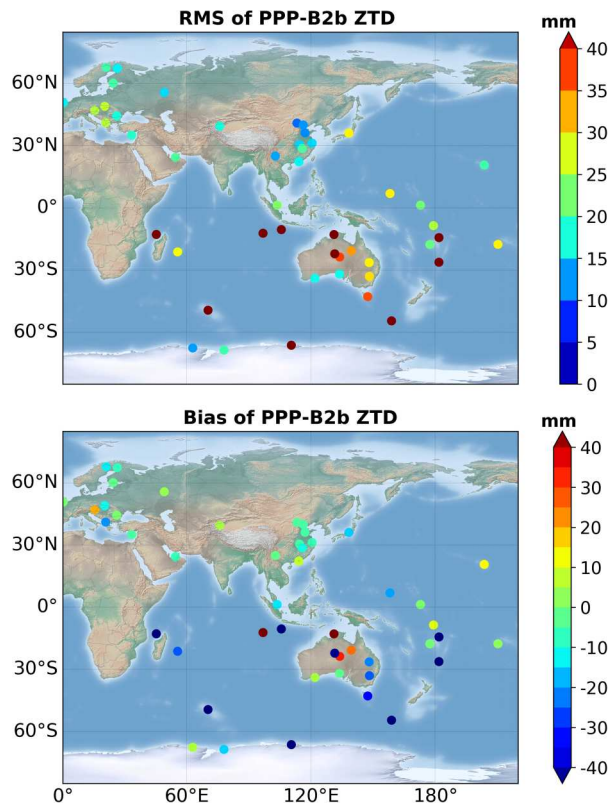
Figure 10 shows the spatial characteristics of the PDOP values, which indicate a property similar to the number of available satellites distributed. According to the results shown in Figure 10, the average PDOP value for satellites with GBM products was mostly within 1–2 with a standard deviation of less than 1. For satellites with PPP-B2b products, only the stations located in East China had STD values comparable to those of the GBM products.

Real-time ZTD and coordinates derived with PPP-B2b using static PPP mode

Using the PPP-B2b corrections and real-time observations, the ZTD and coordinates were calculated in real-time. The results derived with PPP-B2b were evaluated by comparing them with the post-processed results

using the GBM products. Figure 11 shows the RMS and bias of the ZTD derived using PPP-B2b. For the sites located in China, the RMSs of the ZTD are mostly within 10–20 mm, with biases varying from –15 to 15 mm. The lowest RMS was observed at site JNYZ, with an RMS value of 10.57 mm and a bias of –3.95 mm. For the sites located in Europe, the RMSs of the ZTD are mostly within 15–30 mm, with biases varying from –20 to 20 mm. The RMSs of the ZTD for sites located in the Northern Hemisphere were lower than those for sites located in the Southern Hemisphere, which had RMS values larger than 30 mm at most sites. Figure 12 shows the comparison between ZTDs derived using PPP-B2b and GBM orbit/clock products at TAKJ station in China (36.2° N, 117.1° E) and MTIS station in Australia (20.7° S, 139.5° E). For the TAKJ station, located in the central region of the PPP-B2b service, the ZTDs derived from PPP-B2b and GBM show a very good agreement, with differences mainly within ± 20 mm. In contrast, for the MTIS station located outside of the PPP-B2b service region, a significant discrepancy is observed between the two ZTD solutions, indicating noticeably degraded accuracy of the PPP-B2b solution in the Southern Hemisphere.

Figure 13 shows the static position accuracy in the north, east and up directions. The position accuracy for sites located in the Northern Hemisphere was much better than those in the Southern Hemisphere. For the nine stations located in China, the mean RMS errors of the coordinates in the north, east, and up components were 1.1, 1.5, and 2.3 cm, respectively. The biases were mostly



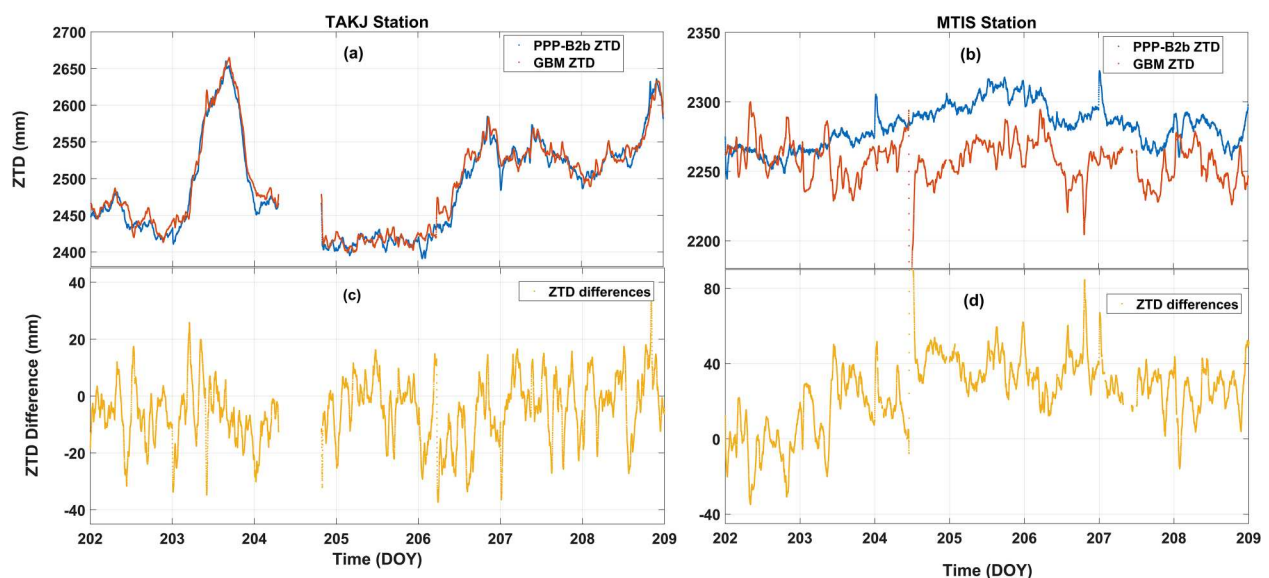
11 RMS (top) and bias (bottom) of the real-time ZTD derived with PPP-B2b products using static PPP mode between DOY 202 and 208 at 48 sites. The ZTD estimated with GBM products was used as the reference. The colour bar shows the RMS or bias of PPP-B2b ZTD with corresponding colours, where values larger than 40 mm are marked in deep red.

within -4 cm to 4 cm. The 3D RMS errors range from 1.2 cm to 8.5 cm, with a mean value of 3.3 cm. For the 11 stations located in Europe and western and central

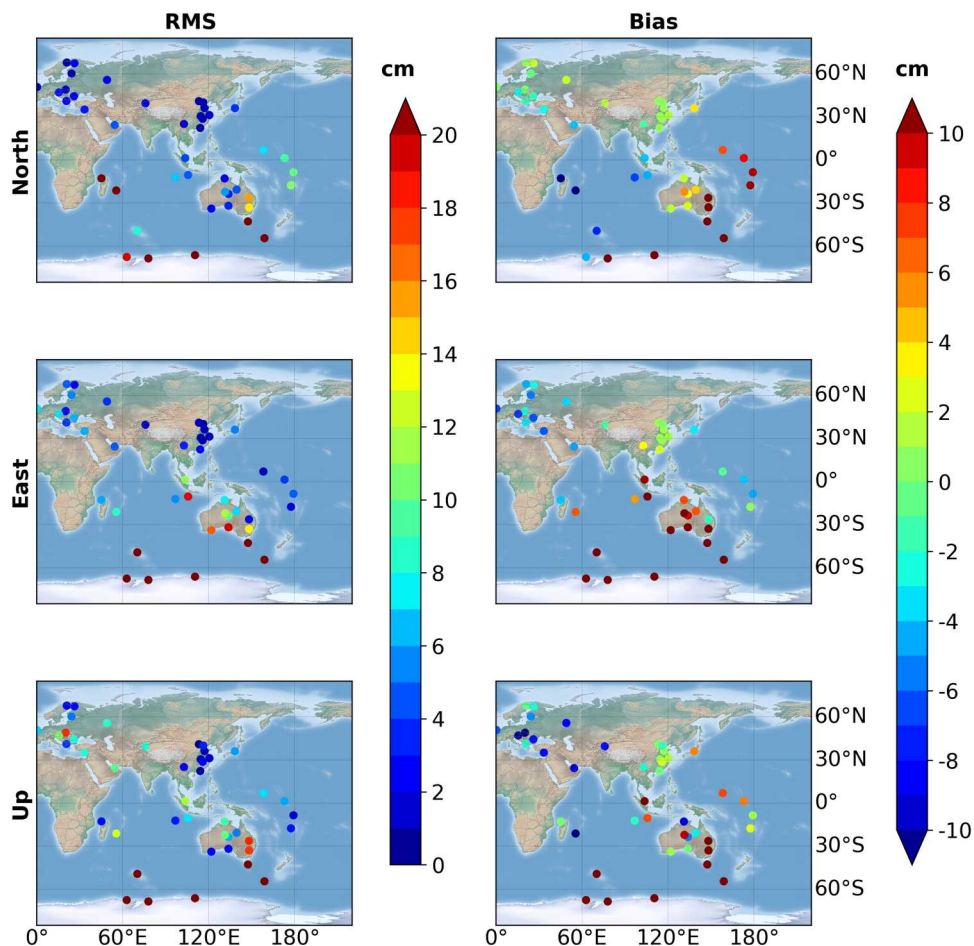
Asia, the mean RMSs of the position errors in the north, east, and up components were 2.1 , 5.0 , and 7.7 cm, respectively. In terms of bias, an obvious negative bias can be observed at most sites in all three components, especially in the up component. The 3D RMS errors range from 4.6 cm to 18.3 cm, with a mean value of 9.8 cm. For sites located in the Southern Hemisphere, the RMS errors are larger than those in the Northern Hemisphere and show a strong dependence on location. For the 9 stations located on the continent of Australia, the mean RMSs of the position errors in the north, east, and up components are 8.9 , 12.6 , and 12.4 cm, respectively, which are much larger than those of sites located in China. Figure 14 shows the comparison between the coordinates estimated using PPP-B2b and GBM at TAKJ and MTIS stations, indicating a slightly better performance at TAKJ station, located within the PPP-B2b service region, than at MTIS station, which is close to the edge of the PPP-B2b service region. The variation in the accuracy of the ZTD estimates and coordinates, which shows much better performance in China compared to Australia or Europe, aligns well with the variation in the number of visible satellites and DOP values across different regions. This agreement is reasonable since the PPP solution is strongly dependent on the number and geometry of the GNSS satellites.

Real-time ZTD and coordinates derived with PPP-B2b using the kinematic PPP mode

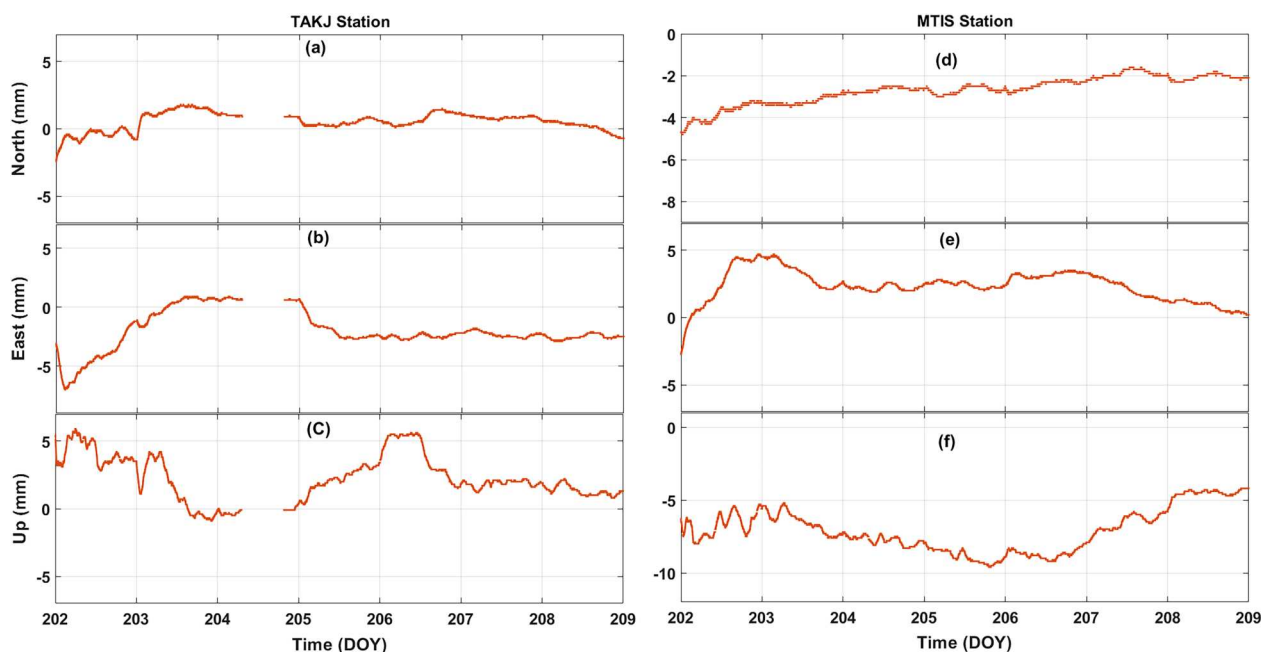
To investigate the performance of the kinematic ZTD estimation with PPP-B2b, an experiment was conducted at 44 global stations between DOY 250 and 256. As shown in Figure 15, the RMS values of the kinematic ZTD values estimated at the eight stations in China and one station in Japan were all within 20 mm, with a mean value of 14.9 mm. The biases are all between -20 and 20 mm. For station ADH1, located in the United Arab Emirates, the RMS value was 24.4 mm, with a bias value of -18.3 mm. At the other stations, the



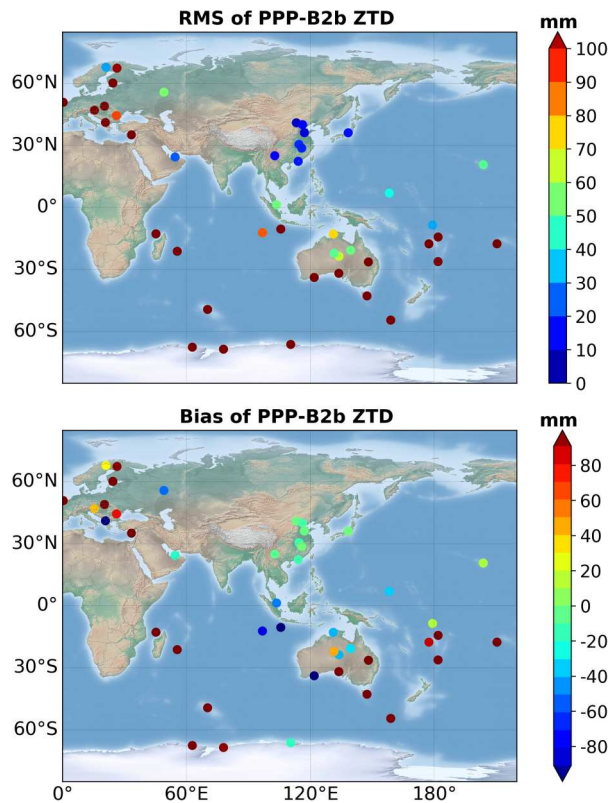
12 Comparison of ZTDs estimated using static PPP mode with PPP-B2b and GBM at stations TAKJ (left) and MTIS (right). The X-axis represents time in DOY, and the Y-axis represents ZTD values or their differences. The top panels show ZTD values for PPP-B2b (red line) and GBM (blue line) solutions, while the bottom panels show their differences



13 RMS (left) and bias (right) of the real-time coordinates derived with PPP-B2b products using static PPP mode between DOY 202 and 208 at 48 sites. The coordinates estimated with GBM products were used as the reference. The colour bar shows the RMS or bias of the PPP-B2b coordinates with corresponding colours, where values larger than 20 cm for the RMS or 10 cm for the bias are marked in deep red and values smaller than -10 cm for the bias are marked in deep blue.



14 Coordinate differences between PPP-B2b and GBM solutions using static PPP mode at stations TAKJ (left) and MTIS (right) in north, east, and up directions.



15 RMS (top) and bias (bottom) of the real-time ZTD derived with PPP-B2b products using kinematic PPP mode between DOY 250 and 256 at 44 sites. The ZTD estimated with GBM products was used as the reference. The colour bar shows the RMS or bias of PPP-B2b ZTD with corresponding colours, where values larger than 100 mm for RMS or 90 mm for bias are marked in deep red, and values smaller than -90 mm for bias are marked in deep blue.

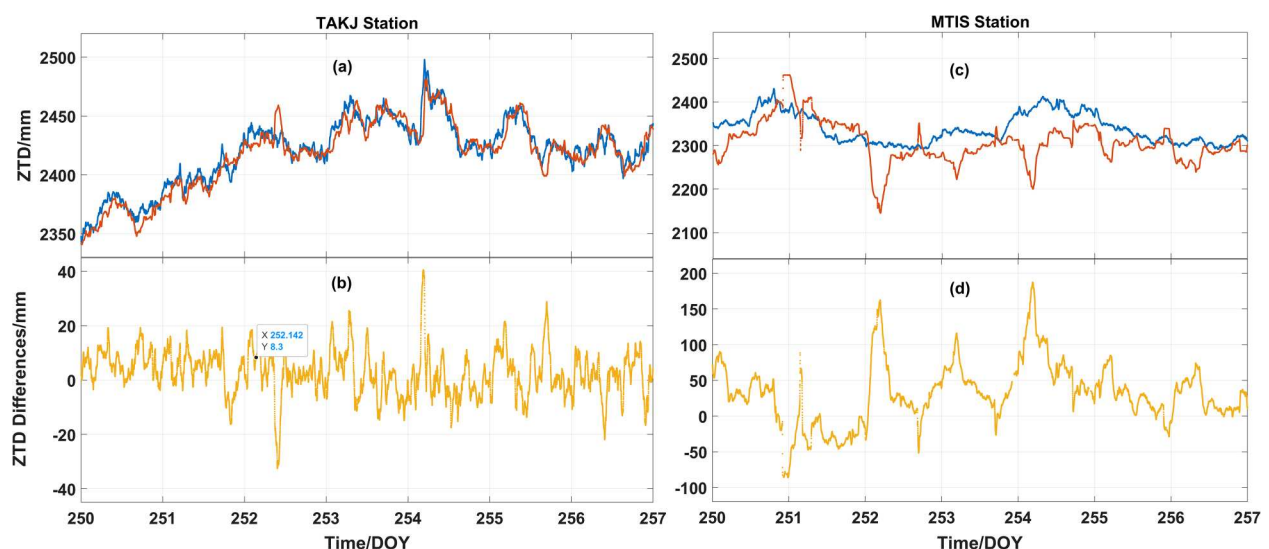
RMS values were greater than 30 mm. This finding indicates that it is only possible to retrieve a high-accuracy ZTD in China and surrounding areas (e.g. Japan) for

the kinematic PPP mode using PPP-B2b information. As shown in Figure 16, the comparison of the time series of ZTDs derived using PPP-B2b and GBM in kinematic mode reveals a strong agreement at the TAKJ station in China, located in the central region of the PPP-B2b service. In contrast, a significant discrepancy is observed at the MTIS station in Australia, located near the edge of the PPP-B2b service region.

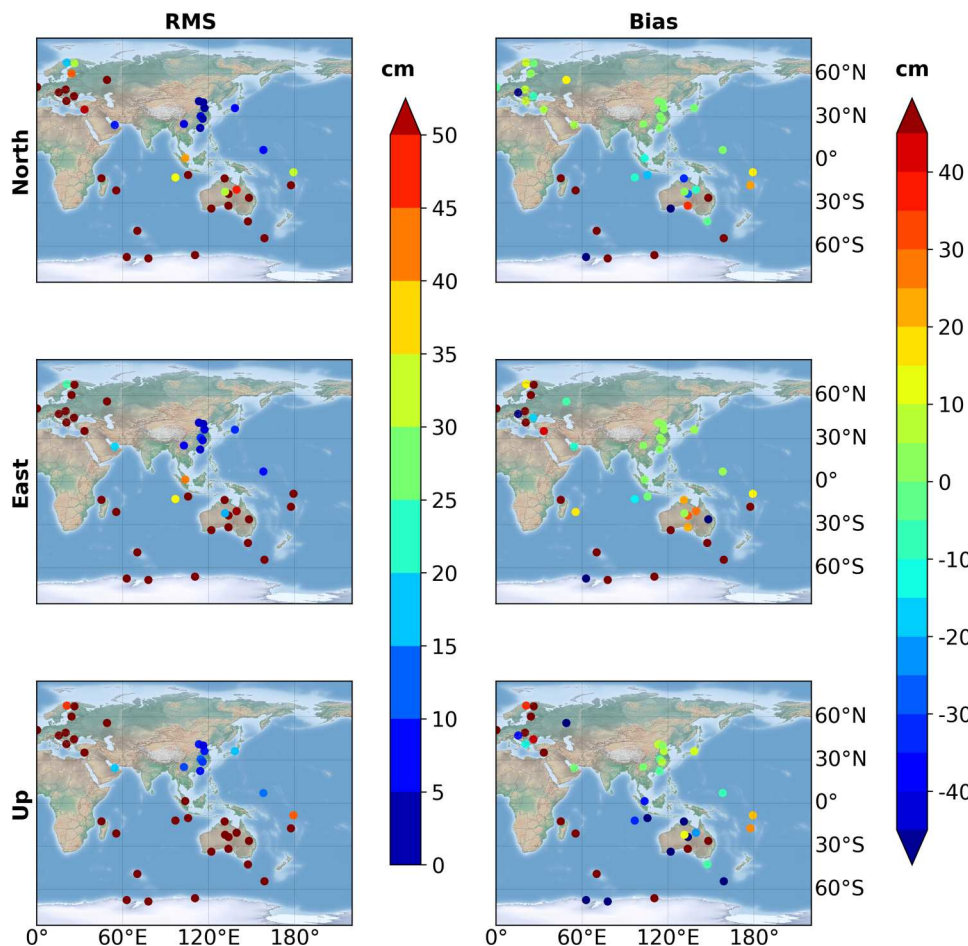
As shown in Figure 17, the spatial properties of the coordinate errors agree well with those of the ZTD error. For the seven stations located in China, the mean RMS errors of coordinates in the north, east and up directions are 2.8, 5.3, and 9.0 cm, respectively, with biases ranging from -4.3 cm to 7.8 cm. The mean 3D RMS error was 10.9 cm across the seven stations. The 3D RMS errors at stations POHN (Micronesia), MSSA (Japan), and ADH1 (Emirates) are 16.8, 20.7, and 27.0 cm, respectively. At other stations, the 3D RMS errors exceeded 50 cm. The time series shown in Figure 18 demonstrates that the coordinate estimation error at the MTIS station is significantly greater than at the TAKJ station. This increased error at MTIS is primarily caused by a lack of visible satellites or poor satellite geometry during certain epochs.

Convergence performance

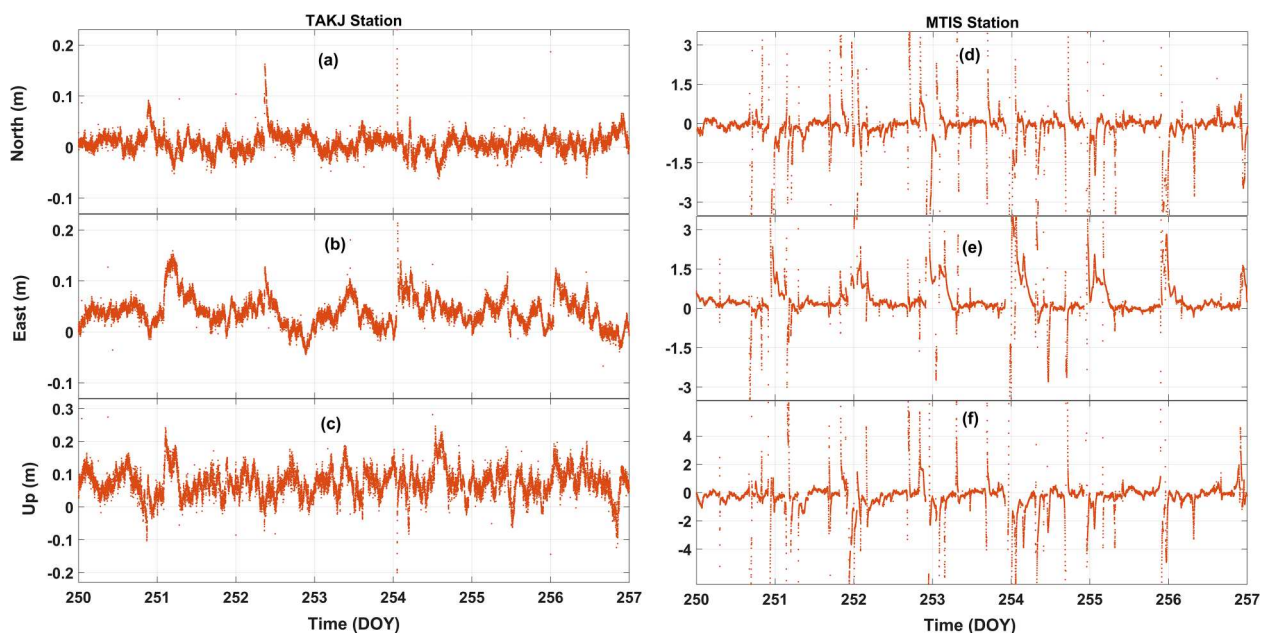
To investigate the convergence performance of the PPP-B2b service in different regions, real-time static and kinematic PPP experiments were conducted on DOY 011 and 015 in 2023 at 45 stations. The static and simulated kinematic PPP processes started at 04:00 UTC on DOY 011 and restarted every 6 h till 08:00 UTC on DOY 015 to obtain 17 convergence periods at each station to study the convergence behaviour in detail. In this study, the PPP convergence time was defined as the time when the position errors were first reached and maintained within 20 (10) and 40 (20) cm in the horizontal (H) and vertical (V) components, respectively, for at least 10 consecutive min. The success rate of convergence and the



16 Comparison between ZTDs estimated using kinematic PPP mode with PPP-B2b and GBM at stations TAKJ (left) and MTIS (right). The X-axis represents time in DOY, and the Y-axis represents ZTD values or their differences. The top panels show ZTD values for PPP-B2b (red line) and GBM (blue line) solutions, while the bottom panels show their differences.



17 RMS (left) and bias (right) of the real-time coordinates derived with PPP-B2b products using kinematic PPP mode between DOY 250 and 256 at 44 sites. The coordinates estimated with GBM products were used as the reference. The colour bar shows the RMS or bias of PPP-B2b coordinates with corresponding colours, where values larger than 50 mm for RMS or 45 mm for bias are marked in deep red, and values smaller than -45 mm for bias are marked in deep blue.



18 Coordinate differences between PPP-B2b and GBM solutions using kinematic PPP mode at stations TAKJ (left) and MTIS (right) in north, east, and up directions.

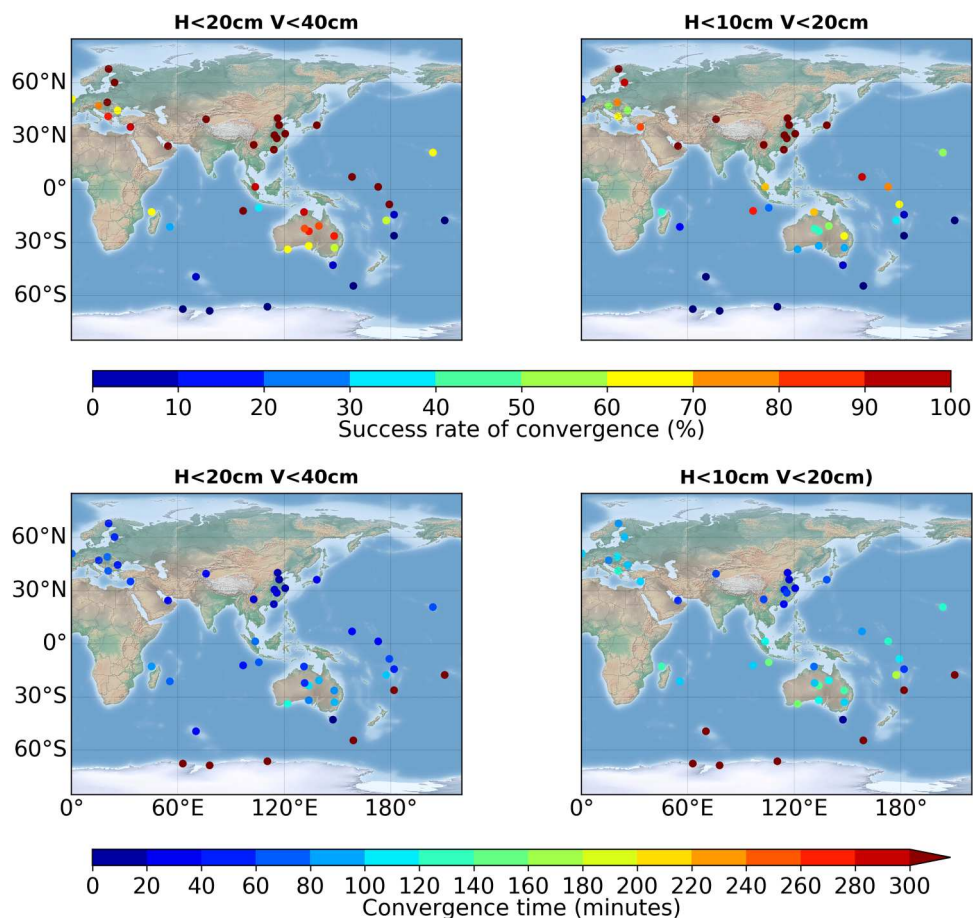
average convergence time for the 17 convergence periods are provided in Figures 19 and 20 for the static and kinematic modes, respectively. As shown in Figure 19, the success rate and average time of convergence of static PPP with PPP-B2b corrections show an obvious dependence on location. The best convergence performance for the PPP-B2b service was observed in China and its surrounding areas. For example, for eight stations located in China, the success rate of convergence was 100% for two convergence criteria ($H < 20$ cm, $V < 40$ cm, $H < 10$ cm, and $V < 20$ cm). The average convergence time for the convergence criterion of $H < 20$ cm and $V < 40$ cm varies from 11.8 min (TAKJ) to 35.9 min (KSKT), whereas for the convergence criterion of $H < 10$ cm and $V < 20$ cm, the convergence time varies from 23.8 min (TAKJ) to 62.6 min (YNTW). For stations far from China, the convergence rate becomes low, which leads to low positioning accuracy, as shown in the previous subsections. The convergence time for the same station exhibited a large variation with epochs. Considering the TAKJ station as an example, the shortest convergence time was only 7.0 min at UTC 04:00 on DOY 013, and the longest convergence time was 55.4 min at UTC 10:00 on DOY 012.

For the simulated kinematic PPP experiments, as shown in Figure 20, the convergence performance exhibited a spatial characteristic similar to that of the static

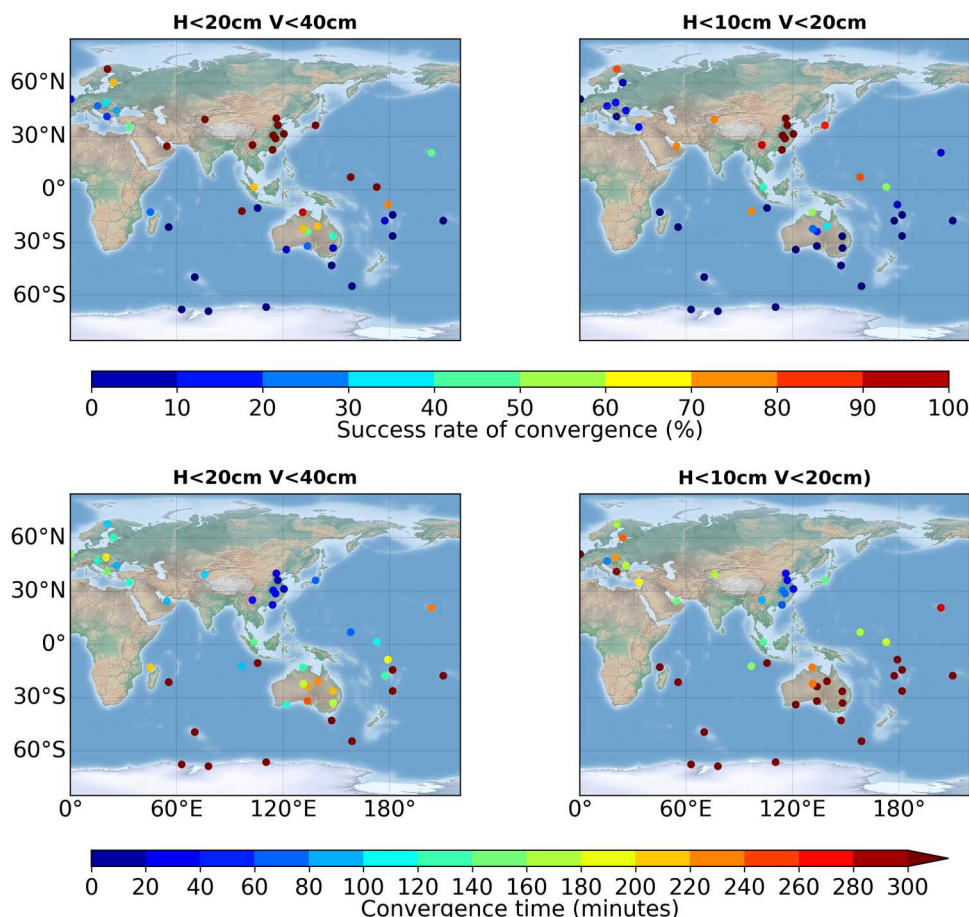
PPP experimental results. The average convergence time of kinematic PPP was longer than that of static PPP. Considering eight stations located in China as an example, the average convergence time varies from 16.2 min (TAKJ) to 100.4 min (KSKT) with the convergence criterion of $H < 20$ cm and $V < 40$ cm and varies from 35.9 min (TAKJ) to 169.4 min (KSKT) with the convergence criterion of $H < 10$ cm and $V < 20$ cm. For station KSKT, which is located in the westernmost city of China, only 13 of 17 experiments successfully converged with the $H < 10$ cm and $V < 20$ cm criteria. For most stations located south of 30° S, the success rate of convergence was below 10% with the criteria of $H < 20$ cm and $V < 40$ cm.

Conclusion

The investigation revealed that the average number of visible satellites varies significantly with location, decreasing steadily from approximately 15 in East China to 4 in the regions south of Australia and Northeast Africa. This led to an increase in the average PDOP from less than 2 in East China to more than 10 in regions far from China, such as South Australia and Africa. An insufficient number of satellites or poor satellite geometry led to the failure of the PPP solution at approximately 50% and 90% of the epochs in South Australia and Antarctica,



19 Success rates of convergence and average convergence times at 45 stations with static PPP. Left panels: the success rate of convergence (left top) and average convergence time (left bottom) with the convergence criterion of a horizontal (H) error smaller than 20 cm and a vertical (V) error smaller than 40 cm. Right panels: the success rate of convergence (right top) and average convergence time (right bottom) with the convergence criterion of a horizontal error smaller than 10 cm and a vertical error smaller than 20 cm. The colour bar shows the success rate of convergence or the convergence time.



20 Success rate of convergence and average convergence time at 45 stations with kinematic PPP. Left panels: the success rate of convergence (left top) and average convergence time (left bottom) with the convergence criterion of horizontal (H) error smaller than 20 cm and vertical (V) error smaller than 40 cm. Right panels: the success rate of convergence (right top) and average convergence time (right bottom) with the convergence criterion of horizontal error smaller than 10 cm and vertical error smaller than 20 cm. The colour bar shows the success rate of convergence or convergence time.

respectively. The accuracy of the positioning and ZTD results agreed well with the spatial variation in the PDOP values of the satellites. For the static PPP experiments, the best results were observed at stations located in China, with ZTD RMS errors mostly within 10–20 mm and 3D coordinate RMS errors of 3.3 cm. The kinematic results revealed that the RMS errors of the kinematic ZTDs estimated at the stations located in China were all within 20 mm, with a mean value of 14.9 mm. For the other stations, the RMS values mostly exceeded 30 mm. In terms of the kinematic positioning results, the mean 3D RMS of the coordinates was 10.9 cm across the seven stations located in China, which is notably better than the performance in other regions. The investigation of convergence performance shows that the success rate of convergence and the average convergence time are highly dependent on the location and time.

As a regional service, the PPP-B2b service performs well in China and surrounding areas (e.g. Japan) in terms of positioning and ZTD estimation but deteriorates significantly in regions far from China (e.g. Australia). Although the Galileo HAS Initial Service was designed as a global service, areas including Eastern Asia, Southeast Asia, Australia, and New Zealand were currently excluded from its coverage. Therefore, a combination of PPP-B2b and HAS services has the potential to provide global

coverage for real-time positioning and ZTD estimation in PPP mode. In this study, the performance of PPP-B2b was evaluated over a one-week period, which is insufficient for assessing its long-term stability. A comprehensive evaluation requires at least one year of study. To facilitate long-term analysis of the PPP-B2b service, we have set up a receiver to receive both PPP-B2b and HAS services, and the received orbit and clock corrections are saved in the SP3 format for future studies.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Appendix

Table A1 shows the detailed information on the GNSS stations used in this study.

Table A1. GNSS station information.

No.	Site	Lat (°)	Lon (°)	City	Country	Tectonic Plate
1	MAC1	−54.5	158.9	Macquarie Island	Australia	Australian/Pacific
2	HOB2	−42.8	147.4	Hobart	Australia	Pacific
3	ESPA	−33.9	121.9	Esperance	Australia	Australian
4	PARK	−33.0	148.3	Parkes	Australia	Australian
5	CEDU	−31.9	133.8	Ceduna	Australia	Australian
6	MCHL	−26.4	148.1	Mitchell	Australia	Australian
7	ALIC	−23.7	133.9	Alice Springs	Australia	Australian
8	MTDN	−22.1	131.5	Mt Doreen Station	Australia	Australian
9	MTIS	−20.7	139.5	Mount Isa	Australia	Australian
10	DARW	−12.8	131.1	Darwin	Australia	Australian
11	COCO	−12.2	96.8	Cocos Island	Australia	Australian
12	XMIS	−10.5	105.7	Christmas Island	Australia	Australian
13	DAV1	−68.6	78.0	Davis	Australian Antarctic Territory	Antarctic
14	MAW1	−67.6	62.9	Mawson	Australian Antarctic Territory	Antarctic
15	CAS1	−66.3	110.5	Casey	Australian Antarctic Territory	Antarctic
16	LAUT	−17.6	177.5	Lautoka	Fiji	Pacific
17	REUN	−21.2	55.6	Le Tampon	France	African
18	FTNA	−14.3	−178.1	Maopo'opo	French Polynesia	Pacific
19	THTG	−17.6	−149.6	Papeete	French Polynesia	Pacific
20	KRGG	−49.4	70.3	Port aux Francais	French Southern Territories	Antarctic
21	MAYG	−12.8	45.3	Dzaoudzi	Mayotte	African
22	OWMG	−26.2	−178.2	Owenga	New Zealand	Pacific
23	TUVA	−8.5	179.2	Funafuti	Tuvalu	Pacific
24	GRAZ	47.1	15.5	Graz	Austria	Eurasian
25	HKSC	22.3	114.1	Hong Kong	China	Eurasian
26	JFNG	30.5	114.5	Wuhan	China	Eurasian
27	YNTW	25.0	102.8	KunMing	China	Eurasian
28	NCEC	28.7	115.8	NanChang	China	Eurasian
29	SKD1	31.3	120.6	SuZhou	China	Eurasian
30	TAKJ	36.2	117.1	Taian	China	Eurasian
31	KSKT	39.5	75.9	Kashgar	China	Eurasian
32	JNYZ	41.0	113.2	Ulanqab	China	Eurasian
33	NICO	35.1	33.4	Nicosia	Cyprus	African
34	POHN	7.0	158.2	Pohnpei	Micronesia	Pacific
35	BJ01	40.1	116.3	Matera	Italy	Eurasian
36	MSSA	36.1	138.4	Misasa	Japan	Eurasian
37	KIRI	1.4	172.9	Betio	Kiribati	Pacific
38	ORID	41.1	20.8	Ohrid	North Macedonia	Eurasian
39	BUCU	44.5	26.1	Bucuresti	Romania	Eurasian
40	KZN2	55.8	49.1	KAZAN	Russian Federation	Eurasian
41	SIN1	1.3	103.7	Singapore	Singapore	Eurasian
42	GANP	49.0	20.3	Ganovce-Poprad	Slovakia	Eurasian
43	KIR8	67.9	21.1	Kiruna	Sweden	Eurasian
44	ADH1	24.4	54.5	Abu Dhabi	United Arab Emirates	Arabian
45	HERS	50.9	0.3	Hailsham	United Kingdom	Eurasian
46	MAO0	20.7	−156.3	Haleakala, Maui	United States of America	Pacific
47	MET3	60.2	24.4	Kirkkonummi	Finland	Eurasian
48	SOD3	67.4	26.4	Sodankyla	Finland	Eurasian